

A Hedonic Analysis of Apple Prices and Product Quality Characteristics in British Columbia

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The quality and market characteristics of apples have important implications for the merchandising strategy of packers and marketers. A hedonic price function relating apple prices to product and market quality characteristics is estimated for British Columbia over three marketing years (1994–96). The results indicate that grade, cultivar, storage and marketing season are the most significant variables influencing apple prices. The results show that price discounts and premiums for quality characteristics are relatively larger for the linear model than for the log-linear or power-transformed models.

Les caractères qualitatifs et commerciaux des pommes ont des répercussions importantes sur les stratégies de vente des emballeurs et des vendeurs. L'auteur examine une fonction hédonique des prix qui relie le prix des pommes aux caractères du produit et à sa qualité commerciale au cours de trois campagnes de mise en marché (1994–1996) en Colombie-Britannique. À partir des résultats, il découle que le classement, le cultivar, la conservation et la période de mise en marché sont les variables les plus significatives du prix des pommes. Les résultats montrent que les rabais et les primes à la qualité sont relativement plus importants selon le modèle linéaire que dans les modèles log linéaires ou dans les modèles à transformée exponentielle.

INTRODUCTION

Quality attributes (cultivar, grade, fruit size) and marketing factors (cold storage, package size, seasonality of sale) are major determinants of apple prices in British Columbia. The large price differentials among apple cultivars suggest that B.C. wholesalers and producers perceive certain cultivars to be associated with desirable grade and quality characteristics (Barichello 1990). In the 1980s, B.C. growers received lower prices than Washington producers, and this was attributed to cultivars grown and the predominance of lower grade products marketed by B.C. Tree Fruits Limited (BCTFL) (Commission of Inquiry–British Columbia Tree Fruit Industry 1990). It has been argued that the high cost structure of production of B.C. apples justifies the need to maintain high quality to secure and expand the market share, coupled with seeking higher prices associated with newer cultivars (Stennes 1994).

Apple pricing policies in B.C. set by packers and BCTFL determine the revenues that growers receive after adjustments for market returns, selling charges and packing house expenses. In some regions of the province, pricing is uniform, but can vary depending on the cultivar marketed, grades, packout and size of the fruit (Calissi 1995). Apple cultivars are like brands in that they identify a subset of a product category, whose members have more characteristics in common with each other than with the members of another category (Richards and Patterson 1998).

The hedonic pricing of apples in Canada is based partly on grades established by the federal government, where grades are influenced by color, maturity and freedom from disease, pests or damage. In addition to grades, hedonic prices are also influenced by other characteristics, including cultivar, storage or fruit size. Federal standards require apples to be “sized” to comply with certain minimum and maximum diameter specifications (www.cfia-acia.agr.ca). A survey in Canada has shown that a wholesaler’s willingness to purchase a particular apple cultivar is influenced by grade, price, taste, appearance and supply availability (British Columbia Ministry of Agriculture, Fisheries and Food 1989). These quality characteristics have become very crucial over the years because of the larger number of apple cultivars in the marketplace and because of retailers’ desire to seek unique traits that are profitable to merchandise.

Hedonic models have been used to derive implicit values of product characteristics for an array of agricultural products, including rice (Brorsen et al 1984), cotton (Ethridge and Davis 1982; Bowman and Ethridge 1992; Chen et al 1997), wool (Angel et al 1990), barley (Wilson 1984), wheat (Veeman 1987; Espinosa and Goodwin 1991), grapes (Golan and Shalit 1993), wine (Oczkowski 1994; Nerlove 1995; Schamel et al 1997), swine (Walburger and Foster 1994), cattle (Parcell et al 1995; Wahl et al 1995) and tomatoes (Bierlen and Grunewald 1995). Fewer hedonic studies (Tronstad et al 1992; Harper and Greene 1993; Kajikawa 1998) specifically look at price–quality relationships of fresh fruits such as apples. Apart from hedonic models, techniques such as conjoint analysis and choice models have been used to examine consumer preferences for apples with differing characteristics (Manalo 1990; Cheng et al 1996). The lack of empirical work related to apple characteristics is somewhat surprising, given the growing importance of newer cultivars and product attributes in the apple marketing system.

While hedonic prices for apples can provide interesting insights into the role of product quality in apple markets, a complete understanding of the relationship between apple prices and other product characteristics including cultivar, storage method, marketing season or fruit size, can provide important information to market traders regarding appropriate marketing strategies to manage inventories, and for assigning priority to factors that augment price premiums. Furthermore, such information can allow plant breeders to assess the importance of key cultivar characteristics for strengthening the competitive position of the apple industry.

This paper estimates a hedonic pricing model and examines the contribution that product quality and market characteristics have on wholesale apple prices. Specifically, we examine the extent to which premiums and discounts are associated with various characteristics in the apple market. Another question addressed is whether the price of apples is explained mainly by grades or whether other product characteristics are of major importance. Unlike Tronstad et al (1992), who address only quality and marketing characteristic variables, we expand the hedonic price analyses to include a variable to adjust for the effect that available supplies can have on apple prices.

BRITISH COLUMBIA APPLE INDUSTRY

Apples are the largest tree fruit crop in B.C. and in 1996 accounted for 90% of the total tree fruit production in the province. About 70% of the apple crop is sold in the fresh market, while the remainder is destined for the processed sector. Since 1961, apple prices have shown large swings and are now on a downward trend (British Columbia Ministry of Agriculture,

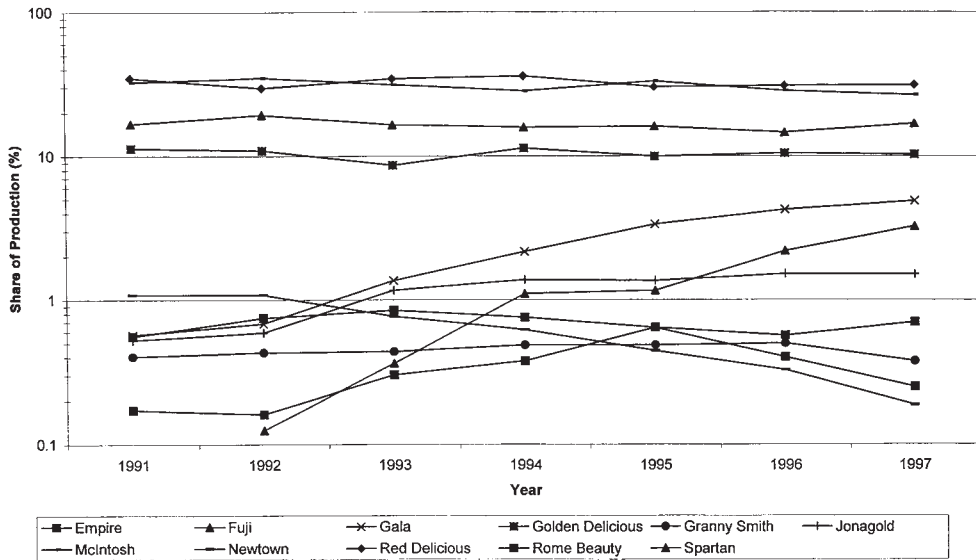


Figure 1. Production shares of selected apple cultivars in the Okanagan, Similkameen, and Kootenay Valleys of British Columbia, 1991–1997.

Source: BC Ministry of Agriculture and Food. Horticultural statistics (various years).

Fisheries and Food. Horticultural Statistics). The majority of apple production is located in the Okanagan, Similkameen and Creston Valleys of the province, and is produced by more than 1,000 apple growers. The major cultivars marketed are Red Delicious, McIntosh, Spartan, Golden Delicious, Royal Gala, Fuji and Jonagold. While Red Delicious remains the predominant cultivar, the production of some of the older cultivars (McIntosh, Spartan, Golden Delicious) has declined in the 1990s. In contrast, Gala, Fuji and Jonagold production have increased rapidly over the past decade (Figure 1). This is a manifestation of structural changes in the industry as consumers appear to be moving away from the traditional varieties and demanding access to a range of newer cultivars noted for their taste and flavorfulness (British Columbia Ministry of Agriculture, Fisheries and Food. Production/Grower Profile 1999).

In B.C., roughly 70% of apples are handled by four cooperatively owned packing houses while independent packers handle the remainder (British Columbia, Ministry of Agriculture, Fisheries and Food, Marketing Profile 1999). Harvested apples delivered to packing houses are sorted, graded and stored in cold or controlled-atmosphere (CA) storage. In some cases, they are packed and sold immediately. Roughly 80% of the apple crop is stored with close to 60% of total storage devoted to CA storage (Deloitte and Touche 1993). Significant improvements in CA storage have increased the storage life of fresh market cultivars such as McIntosh and Golden Delicious. The combined storage capacity available at co-op packing houses throughout B.C. is estimated at roughly 300 million pounds (British Columbia Ministry of Agriculture, Fisheries and Food, Marketing Profile 1999). Storage is necessary to ensure availability and orderly marketing of apples throughout the year.

Apple shipments to packing houses are generally pooled and then graded and priced by cultivar. Apple cultivars such as Red Delicious that are shipped in larger volumes by growers

tend to have lower unit packing costs than some of the newer cultivars, which demand special handling requirements (View West Marketing 1991; British Columbia Ministry of Agriculture, Fisheries and Food, Packinghouse and Processing Profile 1999). Most of the apples produced in the province are marketed by BCTFL, a marketing agency jointly owned by the four packing houses. This agency has the responsibility for market development, securing sales, arranging shipments and coordinating product flow from each of the co-op packing houses (British Columbia Ministry of Agriculture, Fisheries and Food, Packinghouse and Processing Profile 1999). Over the years, BCTFL, acting as a single-desk selling agency, has increased the bargaining power provided to growers and countered the market power of fresh apple buyers in the domestic apple market (Ruggles et al 1991).

Since 1987, a combination of large apple crops and lower retail prices have boosted per capita apple consumption in Canada. Canadian consumption reached 12.76 kilograms in 1994–96, higher than the 1987–91 five-year average of 11.29 kilograms (www.fas.usda.gov). Apples consumed in B.C. are supplied by roughly 16% of B.C.'s total production, with large quantities imported primarily from Washington and southern hemisphere countries.

One of the major issues facing the industry in the late 1980s appears to be the continued decline in the quality of apples available for domestic sale. Two factors that contributed to the decline in product quality are:

- low Canadian prices resulting in the best quality apples being exported to Pacific Rim countries
- inadequate grower returns to compensate for increased production costs, and increased shipping and handling costs stemming from larger grocery retail chains moving toward central warehousing and consolidation of buying functions (Ruggles et al 1991).

To date, the Okanagan Valley Tree Fruit Authority replant program implemented in the early 1990s has boosted improvements in fruit quality resulting in higher packout percentages. While the replant program has improved fruit quality, it has also created logistical problems for packing houses in allocating CA storage space to a more diverse range of products (British Columbia Ministry of Agriculture, Fisheries and Food, Packinghouse and Processing Profile 1999).

CONCEPTUAL FRAMEWORK

Hedonic price functions have been used in the past to derive implicit values of product characteristics for both differentiated agricultural and manufactured products (Berndt 1991). Underlying the hedonic price function is the consumer characteristic model of Ladd and Suvannunt (1976), which is based on Waugh's (1928) earlier work in quantifying the determinants of vegetable prices.

The conceptual model adopted in this study follows Rosen's equilibrium market condition for estimating a hedonic price function. This function can be viewed as representing several equilibrium tangency points for characteristics supplied by producers and demanded by consumers. While the hedonic equilibrium price reveals very little about the underlying supply and demand function for characteristics, it is dependent on the bids that consumers are willing to make for different bundles of characteristics and the offer of those product attributes provided by suppliers.

The buyer's bid function, $\mu = \mu(z_1, z_2, \dots, z_m; g)$ is derived through utility maximization subject to an expenditure constraint, where $\mu(\cdot)$ is the wholesaler's bid function for the

apple market with, $z = (z_1, z_2, \dots, z_m)$ and g corresponding to product characteristics and buyer's expenditure, respectively. The bid function represents the amount that fruit wholesalers are willing to pay for various product characteristics. The partial derivative of the bid function in terms of an individual characteristic depicts the buyer's willingness to pay for an additional unit of characteristic i . However, these values will tend to decrease with an increase in the supply of a particular characteristic.

Similarly, from the supply side of the apple market, a packer or marketing agent offer curve can be described as $\Theta = \Theta(z_1, z_2, \dots, z_m; \delta)$, where $\Theta(\cdot)$ is the packer's offer function, and δ is the supply of the product with given attributes and production technology. The offer function defines the minimum prices that an individual packer is willing to accept for selling δ units of apples with given characteristics z . The partial derivative of the offer function in terms of a given characteristic reveals the seller's marginal implicit value of expanding the supply for a given apple characteristic.

In theory, the determination of apple sales occurs when a particular wholesaler and packer agree on a price for a given apple product with a set of characteristics. This implies that the wholesaler's bid price and the packer offer price are equal, which is associated with the tangency point between the buyer's bid curve and the seller's offer curve for the supply of apple characteristics. At the tangency point, the buyer's marginal implicit bid curve is equivalent to the seller's marginal implicit offer curve for the set of apple characteristics. While individual estimates of the demand and supply functions (Bowman and Ethridge 1992) for characteristics are not the focus of this study, the approach adopted in this paper relies on estimating the market equilibrium price function that comprises the loci of tangency points between the buyer's bid curves and the seller's offer curves.

To estimate the market equilibrium hedonic price function requires obtaining parameter estimates derived by regressing equilibrium apple prices on product quality and marketing characteristics. In the literature, there is some debate regarding the most appropriate functional form to use. Tronstad et al (1992), in their study of the U.S. apple market, adopt a linear model since it is easier to interpret the coefficients. According to Davis (1993), the linear functional form generates the smallest maximum bias if the function is misspecified. The hedonic price function need not be linear, which suggests that the marginal implicit values for product characteristics can be influenced by the level of characteristics purchased by consumers (Rosen; Wahl et al). In this paper, several functional forms are investigated to examine the variation in parameter estimates across functional form specifications.

DATA SOURCES

Weekly transaction data for apples by cultivar, grade, fruit size, package type, storage method, seasonality of sale and shipment by cultivar are obtained from BCTFL (Table 1). In this study, we use a combination of qualitative (dichotomous) and quantitative (continuous) variables. Since most of our variables are qualitative, the partial derivative of price with respect to product characteristics do not represent the traditional implicit product prices, but measure the relative premium or discount over the control variable. Data analysis cover the pooling of three marketing years (1994–96). Apple prices represent FOB prices of apples shipped to wholesalers in Vancouver. Wholesale apple prices are deflated by the Consumer Price Index (1992 = 100) for all commodities in B.C. Tronstad et al (1992) and Veeman (1987) use a similar price index to deflate apple and wheat prices, respectively.

Table 1. Definition of product and market characteristics utilized in the hedonic price function

Variable	Description
Pr_{ij}	Wholesale prices: FOB, Kelowna for a standard carton of apples
Var_{ij}	Apple cultivar: Individual binary variable (0 or 1) value: $v = 1, 2, 3, 4, 5, 6, 7, 8, 9$; 1 = Red Delicious, 2 = McIntosh, 3 = Spartan, 4 = Royal Gala, 5 = Golden Delicious, 6 = Fuji, 7 = Jonagold, 8 = Empire, 9 = other cultivar (Granny Smith, Braeburn, Red Rome, Elstar, Newton, Mutsu); default = Red Delicious
Gr_{ij}	Apple grade: Individual binary variable (0 or 1) value: $g = 1, 2, 3$; 1 = Canada Extra Fancy, 2 = Canada Fancy, 3 = Canada Commercial; default = Canada Extra Fancy
Pk_{ij}	Package size: Binary variable (0 or 1) value: 0 = Tray pack, 1 = Consumer Packages (Carry, Singl, Mini, 3-lb, 5-lb, 8-lb, 10-lb)
St_{ij}	Storage: Binary variable (1 or 0) value: 1 = controlled atmosphere, 0 = regular cold storage
$MKTs_j$	Marketing season: Individual binary variable (0 or 1) value: $s = 1, 2, 3, 4$; 1 = spring, 2 = summer, 3 = fall, 4 = winter; default = winter
Sz_{ij}	Fruit size: Individual binary variable (1 or 0) value: $z = 1, 2, 3$; 1 = large (size < 90), 2 = medium (90 < size < 127), 3 = small (127 < size); default = medium
Q_{in}	Apple shipment: Cultivar (McIntosh, Spartan, Royal Gala, Golden Delicious, Fuji, Jonagold, Empire, other cultivar (Granny Smith, Braeburn, Red Rome, Elstar, Newton, Mutsu)
Interaction terms	grade*cultivar, cultivar*storage, storage*marketing season, marketing season*fruit size

Apart from cultivar prices, the hedonic model includes three distinct grades: Extra Fancy, Fancy and Commercial. As specified by the *Canadian Agricultural Products Act*, apples are graded as either Canada Extra Fancy, Canada Fancy or Canada Commercial. B.C. also has regulations that provide for a B.C. Extra Fancy (blush and stripe) grade for Red Delicious. B.C. Extra Fancy is perceived as a higher-quality grade than Canada Extra Fancy and commands a higher price in the marketplace.¹ Research has shown that “high” grade and “highest” grade apples differ significantly in price and quality. For example, B.C. Extra Fancy returned 15.4 cents per pound to growers versus 11.3 cents per pound for Canada Extra Fancy in the 1986–88 period (Commission of Inquiry–British Columbia Tree Fruit Industry 1990).

The hedonic price model includes apples marketed in different packaging. In the industry, a standard carton of apples represents 42 pounds. This is the most common package employed after apples are graded and sized in the packing houses. For selected niche markets, apples are marketed in consumer packs or polyethylene bags (e.g., 3 pounds, 5 pounds, 10 pounds).

The effects of storage, fruit size, marketing season and apple shipment on prices are also incorporated into the hedonic model. Storage methods include cold and CA storage. After apples are shipped to packing houses, they are either packed for immediate sale or, more commonly, placed into storage. Typically, the better-quality apples are placed into CA storage for sale later in the marketing season. Another quality characteristic included in the model is fruit size. Apple fruit sizes are categorized into small, medium and large. Certain cultivars such as Red or Golden Delicious have a wider range of fruit sizes than McIntosh or Spartan varieties.

Four seasonal dummy variables are included to capture the effects that marketing seasons (fall, winter, spring, summer) have on apple prices. Seasonal market considerations when combined with fruit size, grade and storage are areas of opportunity that offer the most promise for increased returns for U.S. apple producers (Tronstad et al 1992). The model includes a variable to measure apple shipments in order to adjust for the effects that apple supplies have on apple prices. Apple shipments are disaggregated by cultivar to investigate the extent that apple prices are influenced by shipments of individual cultivars.

Several interaction terms are included in the hedonic price function. Traditionally, hedonic models representing the apple market have not addressed potential interdependencies among quality characteristics. For example, in the apple market, possible interdependencies may occur between cultivar and grade or cultivar and storage. Consequently, cultivar may not influence price directly, but may affect price indirectly through its influence on product quality attributes such as grade, storage or fruit size. As a result, returns to growers are influenced not only by cultivar, but also by a cultivar's grade, storage or fruit size.

MODEL SPECIFICATION AND ESTIMATION

The hedonic model relates the deflated wholesale price for a standard carton of apples to several market and product quality characteristics. The joint effects of the characteristics in pairs are included in the model. Previous hedonic price functions for apples and wine have included similar interaction terms (Tronstad et al 1992; Oczkowski 1994). Transformations of price are used to specify the functional form of the model correctly, to provide homogeneity of variance and to make the model easy to interpret. The general hedonic model can be represented as:

$$TP_i = TP_o + \sum Q_j \Phi_{ij} + \sum Q_{jk} \Phi_{ij} \Phi_{ik} + \sum Q_{in} + \epsilon_i \quad (1)$$

where

TP_i = the transformed deflated wholesale price for a standard carton of apples in the i th shipment

TP_o = the base value when prices are transformed

Q_j, Q_{jk} = contributions of the product characteristic j alone or of characteristics j and k in combination to the transformed market value

Φ_{ij} = a dummy variable that is 1 when the i th shipment has characteristic j and zero otherwise

Q_{in} = apple shipment i for cultivar n

ϵ_i = a stochastic error term.

Since Eq. 1 includes only dummy variables to measure qualitative characteristics, the coefficient estimates of the included variables measure the effect of each attribute on price relative to the control variable.

All second-order interactions are considered for inclusion in the model but higher-order interactions are not.² The large number of dummy variables required to do this are generated automatically using software for analysis of variance in SAS (1989). In this analysis of variance, the sources of variation are the explanatory variables (Table 1) and several interaction terms. Each of these sources represents a group of dummy variables. The statistical significance (at the 0.05 level) of each source of variation is used to determine whether the corresponding dummy variables are retained in the model or not. Interaction terms are retained

when the corresponding source of variation is statistically significant. First-order terms are retained when the corresponding source of variation is statistically significant or, to prevent bias in the estimates, when the same quality characteristic is part of a second- order source of variation that is statistically significant.

The relationship between the variances of the residuals and the predicted values is used to determine the power transformation required to produce homogeneity of variance (Fernandez 1992). This analysis shows that the 1/4 power of price would produce homogeneous variances. Preliminary ordinary least squares estimation of the model reveal evidence of autocorrelated residuals. We correct this problem by adopting the PROC AUTOREG procedure in SAS. The Yule-Walker regressor parameter estimates indicate the model correctly models the autoregressive process. The Durbin-Watson statistic reveals that serial correlation is no longer evident in the Yule-Walker corrected model.

Because of the nature of apple marketing, several interaction terms are included in the final model. They are found to be statistically significant (Table 3). Without these interaction terms in the model, much less of the variability in price is explained and homogeneity of variance could not be produced by a power transformation. When there are interaction terms in the model, some quality characteristics are combined with several terms, so there will be collinearity present and the coefficients may not be unique. Dropping a collinear variable may cause biased variance and parameter estimates. Therefore, care must be used in interpreting the coefficients.

Further comparisons are made among models with the untransformed price, (natural) log-transformed price or the 1/4 power of price. Several tests are conducted to determine the most appropriate functional form to represent the hedonic price function. The normality or moments test along with the extended projection (PE) test and Ramsey's specification error test (RESET(2)) (Godfrey et al 1988) are used to determine the model specification for the three transformations of price. The PE test compares a model with a nonnested alternative (H_0 and H_1). The F - value for each comparison using the PE test is statistically significant (Table 2b), but that for the log transformation is smallest. This implies that no transformation provides a perfect specification of the hedonic model, although the log transformation is closest. Godfrey et al suggest that Ramsey's RESET(2) test has good size and power for differentiating between log-linear and linear functional forms under both normality and nonnormality conditions. Ramsey's test requires augmenting the original regressors with powers of the predicted values and testing the significance of the additional variables. All the F -values are again statistically significant, implying that no model is perfectly specified but, again, the log transformation is the closest.

Two reasons may be advanced for the imperfect specification of the model. First, the data are simplified by pooling a number of categories for each quality characteristic (Table 1). While the most important categories are retained, this still means that data for some shipments with distinct and occasionally very different prices are combined. Secondly, interactions among the quality characteristics of order three or higher are not tested, and some of these could have contributed to the price structure. There are tradeoffs between including variables in a single- equation hedonic price function, which are suggested by the dynamics of the apple market and collinearity. Retaining all of the categories in the original data set would be counterproductive, as many infrequently occurring combinations would be included and the complexity of the model (and the number of dummy variables required) would be

Table 2a. Normality (moments) tests

	Model 1	Model 2	Model 3
Skewness	0.0338	-1.2547	-0.7952
Kurtosis	5.3429	8.5458	4.6706

Model 1 – dependent variable price untransformed.

Model 2 – the natural logarithm of price.

Model 3 – the 1/4 power of price.

Table 2b. Non-nested asymptotic model specification PE (extended projection) test

Models	R^2	F -value	P -value
Model 1 vs. model 2	0.6710	396.13	0.0001**
Model 2 vs. model 1	0.6270	139.11	0.0001**
Model 3 vs. model 2	0.6416	212.93	0.0001**
Model 2 vs. model 3	0.6272	148.62	0.0001**

**Denotes highly significant at the ($P < 0.01$) level.

Table 2c. Ramsey's RESET (2) specification error test

Models	R^2	F -value	P -value
Model 1	0.6720	444.04	0.0001**
Model 2	0.6272	150.01	0.0001**
Model 3	0.6416	212.45	0.0001**

**Denotes highly significant at the ($P < 0.01$) level.

substantially increased. Increasing the number of interactions considered would also increase the complexity of the model and, in fact, tax the capacity of the computer system and software. For these reasons, more complex models are not considered. The model with log-transformed price is included because it has the smallest F -values in the specification tests, because it is sufficiently similar to the 1/4 power transformation that it introduces little heterogeneity of variance and because coefficients from log-linear models have a clear economic meaning.

RESULTS AND DISCUSSION

Table 3 provides Yule-Walker coefficient estimates for the three estimated models. The variables explain over 70% of the variation in apple prices, with F -values suggesting that grade, cultivar, storage and marketing season are the principal factors influencing apple prices. The estimated models include all the variables in Table 1 with the exception of Red Delicious apple shipments. This variable is found to be consistently not significant and is omitted from

Table 3. Hedonic (Yule-Walker) estimates of product quality attributes and market characteristics effects on apple prices

	Model 1			Model 2			Model 3		
	Coefficient	T-statistic	F-value	Coefficient	T-statistic	F-value	Coefficient	T-statistic	F-value
Grade	Intercept	15.1264	78.24	2.6873	226.59	5193.00**	1.9612	332.51	5042.58**
	Extra Fancy (control)	-2.6009	-9.14	-0.1643	-9.59		-0.0818	-9.51	
	Fancy	-6.5505	-14.96	-0.4812	-18.15		-0.2302	-17.34	
Cultivar	Commercial					1227.64**			1424.84**
	Red Delicious (control)								
	Empire	2.1739	4.86	0.1407	5.22		0.07	5.17	
	Fuji	10.8018	13.44	0.5958	12.06		0.306	12.46	
	Golden Delicious	0.4474	1.62	0.0258	1.54		0.0134	1.60	
	Jonagold	3.6361	8.28	0.2078	7.80		0.1067	8.01	
	McIntosh	0.8916	2.88	0.0512	2.73		0.0266	2.84	
	Royal Gala	14.6491	33.81	0.7033	26.66		0.3736	28.38	
	Spartan	3.0426	10.96	0.1734	10.26		0.0887	10.52	
	Other Cultivar	0.0504	0.14	0.0005	0.02		0.0009	0.09	
Storage	Regular Storage (control)					35.63**			533.82**
	Controlled Atmosphere (CA)	3.5700	23.76	0.2426	25.37		0.1186	25.40	
Package	Tray (control)					266.16**			273.46**
	Consumer Pack (e.g., 5-lb bags)	-0.1143	-1.56	-0.0125	-2.68		-0.0057	-2.52	
Fruit size	Medium: 90 < size < 127 (control)								
	Large: size < 90	-0.2736	-3.65	-0.0227	-4.72		-0.0104	-4.46	
	Small: 127 < size	-1.5637	-19.08	-0.0908	-17.33		-0.0458	-18.00	
Marketing season	Winter (control)					482.10**			482.00**
	Fall	1.87	13.61	0.1341	15.58		0.0644	15.21	
	Spring	-2.2647	-7.40	-0.1704	-8.76		-0.0810	-8.52	
Cultivar*grade	Summer	2.4915	9.84	0.1772	11.07		0.0854	10.89	
	Fuji*Commercial	-5.1538	-4.23	-0.1959	-2.54		-0.1129	-2.99	
	Golden Delicious*Commercial	-0.4406	-0.56	-0.0549	-1.15		-0.0232	-0.98	
	McIntosh*Commercial	-0.6480	-1.11	-0.0333	-0.94		-0.0178	-1.01	
	Royal Gala*Commercial	-12.6333	-13.28	-0.6622	-11.27		-0.3412	-11.71	
	Spartan*Commercial	-1.3215	-2.21	-0.0458	-1.27		-0.0277	-1.53	
	Jonagold*Commercial	-2.7430	-2.46	-0.1137	-1.62		-0.0651	-1.89	
	Other Cultivar*Commercial	1.8747	1.56	0.1745	2.28		0.0790	2.11	
	Golden Delicious*Fancy	-0.2385	0.52	-0.0131	-0.48		-0.0068	-0.49	
	Spartan*Fancy	-1.3223	-3.39	-0.0574	-2.44		-0.0315	-2.67	
	Other Cultivar*Fancy	2.8236	5.01	0.1775	5.20		0.0883	5.17	

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Table 3. Continued

Variable	Model 1			Model 2			Model 3		
	Coefficient	T-statistic	F-value	Coefficient	T-statistic	F-value	Coefficient	T-statistic	F-value
Cultivar*storage	Fuji*CA	-7.8017	-9.17	40.05**	-0.4267	-7.87	-0.2201	-8.31	33.16**
	Golden Delicious*CA	-0.2588	-1.14		-0.0160	-1.12	-0.0084	-1.20	
	Jonagold*CA	1.4083	3.42		0.0787	3.02	0.0397	3.11	
	McIntosh*CA	-0.2215	-0.98		-0.0091	-0.64	-0.0058	-0.83	
	Royal Gala*CA	-2.544	-6.48		-0.1194	-4.82	-0.0639	-5.26	
	Spartan*CA	-0.7417	-3.50		-0.0532	-3.98	-0.0262	-4.00	
Storage*marketing season	Other Cultivar*CA	2.4013	6.68		0.1545	6.68	0.0755	6.72	
	CA*fall	-2.3226	-5.27	193.27**	-0.1681	-5.99	-0.0788	-5.77	225.44**
	CA*spring	1.4883	4.80		0.1179	5.98	0.0554	5.77	
Fruit size*marketing season	CA*summer	-4.0742	-14.06		-0.2856	-15.51	-0.1379	-15.32	
	Large*fall	0.4902	4.30	7.92**	0.0257	3.51	0.0132	3.72	8.25**
	Small*fall	0.0977	0.87		0.0067	0.94	0.0031	0.90	
	Large*spring	-0.4756	-4.32		-0.0257	-3.64	-0.0135	-3.95	
	Small*spring	-0.3562	-3.22		-0.0239	-3.39	-0.0118	-3.45	
	Large*summer	-0.4959	-2.53		-0.0363	-2.88	-0.0176	-2.88	
Shipment size	Small*summer	-0.1228	-0.76		-0.0162	-1.58	-0.0072	-1.45	
	McIntosh	-0.0001	-0.15	25.79**	-0.0001	-0.15	-0.0001	-0.17	24.17**
	Spartan	-0.0001	-0.44	9.09**	0.0001	0.32	0.0001	0.12	12.41**
	Golden Delicious	-0.0001	-1.43	46.04**	-0.0001	-1.71	-0.0001	-1.64	47.36**
	Royal Gala	-0.0001	-0.59	8.10**	0.0001	0.77	0.0001	0.45	4.62*
	Empire	0.0005	2.76	102.97**	0.0001	2.86	0.0001	2.84	94.66**
Number of observations = 15,574	Jonagold	0.0002	0.81	14.62**	0.0001	0.94	0.0001	0.90	14.05**
	Fuji	-0.0024	-4.22	94.16**	-0.0002	-4.59	-0.0001	-4.51	100.14**
	Other Cultivar	-0.0004	-2.76	30.58**	-0.0001	-2.68	-0.0001	-2.72	30.04**
D-W	2.14			2.13			2.13		
R ²	0.7514			0.7162			0.7295		

Notes: Model 1 – linear, model 2 – log-linear and model 3 – power transformation of price.

*Significant ($P < 0.05$); ** highly significant ($P < 0.01$).

subsequent analyses. The constant term in the regression models represent the joint effect of the default values. The combination of a relatively high R^2 and a large number of explanatory variables that are significant at the 1% level suggest that product quality variables and their interactions explain a significant amount of the variation in apple prices.

As expected, apples graded Canada Commercial or Canada Fancy are discounted relative to those graded Canada Extra Fancy.³ The estimated parameters are significantly different from zero, indicating that grades are an important variable shaping apple prices. Price discounts for Commercial and Fancy grade apples are more pronounced in model 1 than they are in models 2 and 3. The depth of price discounts for Commercial grade apples is much greater for newer cultivars than for older ones. In model 1, price discounts for Commercial grade apples range from \$12.63 per carton for Royal Gala to \$2.74 per carton for Jonagold. Fruit buyers of Commercial grade Spartan are paying \$1.32 per carton less than for Extra Fancy Red Delicious. The results from model 2 show that price discounts for a Commercial grade Royal Gala and Fuji apple are \$0.66 per carton and \$0.20 per carton less than for Extra Fancy Red Delicious.

It is widely known that the industry grading standards are based primarily on color and the aesthetic appearance of the fruit. However, over the past 10 years, consumers have been demanding newer cultivars that are tastier, flavorful, but not redder and more shapely than some of the older cultivars. Therefore, the grading standards ought to be revised to reflect changing consumer preferences for newer cultivars and incorporate other elements (e.g., soluble solids) into the grading system.

While the acreage in newer cultivars has increased over the past 10 years, our results reveal that Royal Gala, Fuji and Jonagold receive a substantial price premium over Red Delicious. Empire and Spartan, a couple of older cultivars also enjoy a price premium, though the magnitude of the price increase is smaller than for the newer cultivars. Table 3 shows that for model 1, Royal Gala, Fuji and Jonagold earned a price premium of \$14.65, \$10.82 and \$3.64 per carton above Red Delicious apples. Spartan and Empire also enjoy a price premium of \$3.04 and \$2.17 per carton, respectively. The level of price premium for newer and older cultivars relative to Red Delicious is much smaller for models 2 and 3 than they are for model 1. These results suggest that larger price premiums do exist for newer cultivars and therefore marketing efforts need to target fruit buyers and retailers to ensure they are aware of novel cultivars and their quality characteristics. Findings from a recent consumer survey reveal that the apple variety most purchased by Canadian consumers is McIntosh, followed by Granny Smith, Golden Delicious, Spartan and Red Delicious (Benchmark Research 1998). While these cultivars may be profitable for retailers and have good aesthetic appeal on the retail shelf, they do not rank high in fruit quality relative to the newer cultivars. Research by the IMPACT Center in Washington State suggests that price premiums for Jonagold, Fuji and Gala cultivars may still be available for another two to three years (O'Rourke 1999).

A priori, one would expect a higher proportion of CA fruit to fetch price premiums than regular stored fruit, since CA fruit is generally firmer and considered to be of higher quality. Some of the older cultivars such as Red Delicious tend to store better in CA than the newer cultivars including Gala and Fuji. CA fruit provides the co-op packing houses with a competitive edge over most of the smaller independents in terms of market timing and a year-round supply of fruit. It is shown from Table 3 that apples in CA benefit from a price premium relative to regular stored apples. The CA coefficient is significantly different from zero at the 1%

level, implying that a high level of confidence can be placed on this variable. Of the cultivars in CA, Jonagold and "other cultivars" are the only ones that report price premiums above a Red Delicious cold-stored apple. Jonagold is one of the few high-priced apples that has to be in CA instead of regular storage to be marketed year-round. The results of model 1 suggest that the price premium for Jonagold and "other cultivars" in CA is \$1.41 and \$2.40 per carton above cold-stored Red Delicious. In contrast, large price discounts are reported for Fuji and Royal Gala in CA, \$7.80 and \$2.54 per carton, respectively. This result may be due to Gala having a relatively short storage life and not being prone to many of the cosmetic disorders that affect other cultivars in CA. Fruit buyers' demand for CA apples appears to be much stronger in the spring than either the fall or summer. Model 1 results indicate that CA fruit in the spring fetches a price premium of \$1.48 per carton above cold-stored apples sold in the winter. This premium is much larger than the price increases of \$0.12 and \$0.06 per carton obtained from models 2 and 3. Higher price premiums for CA fruit in the spring may be due to the limited availability of apple supplies in the spring months coupled with greater consumer demand. Recent changes in CA storage standards (180 days down to 45 days) has provided CA fruit the opportunity to compete with regular stored fruit as early as December (British Columbia Ministry of Agriculture and Food, Packinghouse and Processing Profile 1999).

The decision to place apples in CA storage is influenced by the storage potential of fruit, marketing considerations and the timing of sales. CA storage decisions are also influenced by the level of imports. Most apple imports are provided by Washington State suppliers (Agriculture and Agri-food Canada, Trade Statistics). With increased competition from imports, seasonal price differences will continue to decrease, thereby reducing the profitability of CA investments.

How apples are packaged, size and the season in which they are marketed do have a significant effect on apple prices. Apples packaged in polyethylene packs such as three- or five-pound bags are discounted relative to apples merchandised in the conventional tray packs (Table 3). The coefficient on consumer packs is statistically significant at the 1% level for both model 2 and model 3, implying that it is a significant factor affecting apple prices in these models. In general, apples marketed in consumer packages are designated for consumer convenience and in most cases are smaller and lower-graded fruit.

Closely related to packaging is fruit size. In practice, one would expect larger apples to fetch higher price premiums over smaller or medium-sized fruit since the reward structure tends to favor larger size and redder apples. However, wholesale fruit buyers are willing to pay higher prices for medium-sized apples over smaller or larger ones. This is supported by a survey that showed 73% of Canadian households prefer a medium-sized apple (Benchmark Research 1998). The results from model 1 reveal that small and large apples relative to medium-sized ones receive a price discount of \$1.56 and \$0.27 per carton, respectively. The lower price discount for larger apples may be due to lower firmness from storage. Stuible and Kennedy (1985) suggest that size is a more significant factor shaping apple prices than sales timing, and therefore the production of larger apples by expensive management practices such as thinning may be more beneficial.

There are marked differences pertaining to the marketing seasons, with the fall and summer seasons providing greater opportunities to earn higher apple prices than the spring and winter months. Price premiums earned in the summer and fall are \$2.49 per carton and \$1.87 per carton above winter prices based on the results from model 1. They are higher than the

premiums of \$0.18 and \$0.13 per carton earned in the summer and fall months from model 2. The cyclical nature of apple prices is closely associated with the marketing season. Because of climate and maturity differences among cultivars, the harvest season extends from August through October, with McIntosh, Gala and Jonagold harvested in the earlier part of the season followed by Red Delicious, Fuji and Granny Smith in the latter half. During the months from April through July, domestic apple prices are relatively weaker. This is evident by the lower prices obtained for small and large fruit sold in the spring months relative to medium-sized fruit marketed in the winter. Price discounts for small and large fruit merchandised in the spring range from \$0.36 per carton to \$0.48 per carton for model 1. Larger apples merchandised in the fall season tend to reap a higher price premium (Table 3).

Irrespective of the model, the majority of coefficients on the individual cultivar shipment variables are not statistically significant at the 1% level. This implies that apple cultivar shipment is not a major factor influencing apple prices. Similar results are reported by Kajikawa (1998) for the Japanese apple market. Table 3 shows that Fuji, Empire and "other cultivars" are the only coefficients that have *t*-values that are significantly different from zero. This may be partly due to the average annual percentage increases in production reported for Fuji and Empire during the sample period. Fuji production increased by roughly 8.5 million pounds, while Empire jumped by 1.6 million pounds from 1991 to 1997. Over the years, the production of Red Delicious, McIntosh and Spartan has declined, while the production of newer cultivars has increased (British Columbia Ministry of Agriculture and Food, Production/Grower Profile 1999).

CONCLUSION

This study employs a hedonic price function to estimate price discounts and premiums associated with quality and market characteristics for apples marketed in British Columbia. Product characteristics considered include cultivar, grade and fruit size, while marketing attributes are storage, packaging, seasonality of sale and shipment size. The study concludes that grade is one of the principal characteristics influencing apple prices, with cultivar, storage, marketing season and fruit size explaining a relatively significant proportion of prices. Interactions such as cultivar*grade and storage*marketing season illustrate the importance of interdependencies among fruit characteristics in the apple market.

The results indicate that wholesale fruit buyers are willing to pay a premium price for some of the newer cultivars including Fuji, Royal Gala and Jonagold. Consequently, producer incentives should be provided to speed the replacement of traditional cultivars by newer varieties. This would not only create opportunities for domestic expansion in high-priced cultivars but could increase consumer preferences for tastier and flavorful apples such as Gala and Fuji. To match product quality with consumer preferences for newer cultivars would require providing orchardists with the incentive to grow larger and better quality apples. This would require modifying the grading standards and placing greater emphasis on fruit pressure tests and soluble solids instead of rewarding growers for the reddest and largest apples. These changes in marketing may require packing houses to avoid offering discounts for volume sales and instead promote the quality of the fruit and the service provided.

In this study, we find that considering cultivar in combination with grades or storage can provide marketers with important price policy information. Firstly, it allows one to know the difference in price effects between older and newer cultivars that grade Commercial or Fancy.

Secondly, it illustrates the magnitude of price discounts or price premiums associated with individual cultivars in CA. Considering that CA is a very expensive capital cost item, it may be more prudent to allocate more CA storage to high-priced cultivars (e.g., Jonagold) that can maintain good fruit quality and compete with imports from southern hemisphere countries.

In this study, we treat apples as a generic commodity. An alternative approach would be to estimate a hedonic price function separately for newer and traditional cultivars and to determine how price discounts and price premiums for various product characteristics may differ between these groups.

NOTES

¹A preliminary analysis of the raw data indicates several gradations of Extra Fancy and Fancy for Red Delicious based on a given proportion of red color. From a marketing efficiency perspective, the range of what is designated Canadian or B.C. Extra Fancy may appear too wide. Raising the minimum color requirement in the grading system could eliminate the need for additional Extra Fancy and Fancy grades and provide an opportunity for another category to be developed.

²Product characteristics ought to be truncated to include only distinct quality attributes since a high degree of collinearity between explanatory variables can prevent estimation of distinct effects (Ethridge and Davis 1982).

³It is apparent that the lower number of apples graded Canada Extra Fancy may be related to the average age of orchards. Calissi (1995) finds that older apple orchards that are typical of the Okanagan in the late 1980s contributed to the low percentage packout of high-quality fruit.

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